

HARSH GUPTA

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EDUCATION

Georgia Institute of Technology

Master of Science in Analytics (Computational Data Analytics Track)

Aug 2025 – Dec 2026 (Expected)

Atlanta, USA

Indian Institute of Technology Roorkee

Bachelor of Technology in Chemical Engineering

Jul 2015 – May 2019

Roorkee, India

EXPERIENCE

Athena

Machine Learning Engineer Intern

May 2026 – Present

Atlanta, USA

- Built the LLM labeling layer of a production churn-prediction pipeline on the **OpenAI Batch API** with **GPT-5**, classifying warning and positive signals from client-call documents into a **42-category risk taxonomy**.
- Developed an explainable logistic-regression churn model on those categorized signals among others, improving on a **~2.3K-feature XGBoost** with **143** interpretable features and lifting **PR-AUC from 0.45 to 0.49**.
- Shipped the chain end to end as a fully **incremental** pipeline on **Prefect** and **AWS ECS/Fargate**, with per-stage anti-joins on deterministic hash keys making every stage idempotent and restartable.
- Migrated labeling onto **Amazon Bedrock** batch inference, swapping GPT-5 for **Amazon Nova 2 Lite** on a **500-sample evaluation** of label agreement and cost.
- Partnered with **partnership ops** to build a churn **attribution** framework across 9.9K client partnerships in **Snowflake**, correcting **1,150 misattributed churn events** and surfacing a **\$13.5M ARR** blind spot.

FUJIFILM Group

Senior Data Scientist

Nov 2019 – Jun 2025

Tokyo, Japan

- Built a **RAG** assistant over 500+ repair SOPs with **Azure AI Search** hybrid retrieval and **GPT-4**, reaching **0.88 recall@5** on a 120-question eval set and returning cited answers or abstaining when evidence was missing.
- Fine-tuned **BERT** for multi-class routing of 100K+ support tickets with class-imbalance handling and **confidence thresholding**, auto-routing **71% at 94% precision** and cutting misroutes from 18% to 6%.
- Led a team of 5** to build and deploy a **CatBoost** failure-prediction pipeline for endoscopes: **91% recall** and 28% fewer unexpected device failures.
- Developed an **XGBoost** contract pricing simulator ($R^2 = 0.94$) that surfaced pricing and margin levers, generating **\$1.2M in incremental revenue**.
- Partnered with repair workshop **SMEs** to design and analyze **A/B tests** on alternative repair countermeasures, identifying the top-performing approach and cutting failure recurrence by **15%**.
- Consolidated JP-EU-US contract analytics into 15+ **Azure Databricks** ETL pipelines and standardized **Power BI** dashboards for cross-functional teams, saving **150 person-hours/year**.

PROJECTS

MedeFence – Multi-Agent Red-Team Testbed for LLM Agent Safety | *Python, LLMs, Evals*

Aug 2026

First-author paper under review, NeurIPS 2026 IAEval Workshop

- Built a two-agent red-team/blue-team testbed (Llama-3.3-70B attacker, Qwen-2.5-72B target) with deterministic finite-state-machine scoring: **1,281 trials** across 5 strategies, a **2×3 factorial** of tool defenses, and a monitor arm.
- Showed safety metrics track *where* a guardrail intercepts, not how the agent behaves: a tool gate scores a definitional 0.00 against an output monitor's deceptive 0.02; preserved counterfactuals recover a **0.91 interception rate**.

CanvasAI – LLM-Powered Agentic Canvas Assistant | *Python, GenAI, Tool Calling*

Feb 2026

`canvas-gpt.streamlit.app`

- Built an agentic assistant for Canvas LMS answering natural-language questions on assignments, grades and course content, and auto-syncing deadlines through the **Google Calendar API**.
- Multi-provider LLM support (OpenAI, Anthropic, Groq) over a **tool-calling** framework, deployed as a conversational app on **Streamlit**.

LLM-Based Music Generation (MIDI) | *Python, PyTorch, LLMs, Fine-Tuning*

Dec 2025

- Fine-tuned **GPT-2** and **T5** on 45K+ MIDI files with REMI+ tokenization for music continuation and span infilling.

TECHNICAL SKILLS

LLM & GenAI: Large Language Models, Generative AI, Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation, Prompt Engineering, Tool Calling, LLM Evaluation & Red-Teaming, Fine-Tuning, Batch Inference

Languages & Tools: Python, SQL, C++, Git, Docker, Terraform, AWS (Bedrock, ECS, S3), Snowflake, Databricks, Azure ML, Prefect, PySpark

Frameworks & ML: PyTorch, Hugging Face Transformers, LangChain, scikit-learn, XGBoost, CatBoost, LightGBM, MLOps, Model Deployment, Feature Engineering, Experimentation

LEADERSHIP & AWARDS

- Founded **Data Sprout**, an online community of 10K+ data learners; conducted 11 workshops on data science.
- Awards: Recipient of the company-wide **FUJIFILM Software Team Award 2023** | **Kaggle Notebooks Expert**